SINGAPORE DIMENSIONS • GRADE 3A • CHAPTER 5

Multi-Step Word Problems & Bar Models

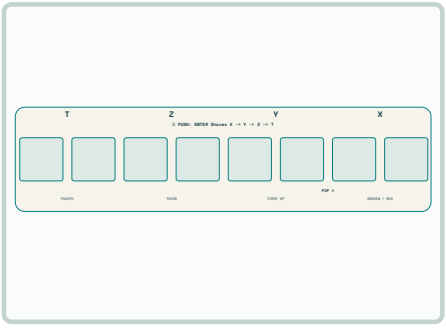
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.OA.D.8

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Stack Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Build tape models with cube blocks: 24 total, subtract 6, split into 3.

Sketch Model on Paper:
Draw Singapore bar model: total bar with bracket and partition.

3. Verify with SC-6 (Abstract)

RPN Stack Register Flow:
Automatic stack lift retains difference in Y; entering 3 lifts to X; [÷] solves without parens.

Self-Check Keystroke Sequence:

`2` `4` `enter` `6` `-` `3` `÷`

Expected LCD: 6 stickers each

1 **Sarah has $50, buys 2 shirts at $15 each. Change?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

2 **A baker bakes 40 loaves, sells 16, packs remainder in 4 bins.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

3 **5 boxes with 8 pencils each, plus 12 extra loose pencils.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

4 **Three friends each buy a $4 snack, and split a $9 pizza. Total cost?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 5

Multi-Step Word Problems & Bar Models • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Interlocking Cubes & 4-Line Hardware LCD Stack

CAD Blueprint: stack_board

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/build/scad/stack_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal 4-Register Rail & White Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Sarah has $50, buys 2 shirts at $15 each. Ch** Standard: CCSS.MATH.3.OA.D.8	**$20**	[1][5] [ENTER] [2] [×] [5][0] [ENTER]
2	**A baker bakes 40 loaves, sells 16, packs rem** Standard: CCSS.MATH.3.OA.D.8	**6 loaves per bin**	[4][0] [ENTER] [1][6] [-] [4] [÷]
3	**5 boxes with 8 pencils each, plus 12 extra l** Standard: CCSS.MATH.3.OA.D.8	**52 pencils**	[5] [ENTER] [8] [×] [1][2] [+]
4	**Three friends each buy a $4 snack, and split** Standard: CCSS.MATH.3.OA.D.8	**$21**	[3] [ENTER] [4] [×] [9] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Traditional calculators require (24 - 6) / 3; RPN evaluates naturally in sequence.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.